

Kathryn D. Huff

CONTACT INFORMATION	Associate Professor Dept. of Nuclear, Plasma, and Radiological Engineering University of Illinois at Urbana-Champaign	Mobile: (281) 734-1342 UIUC e-mail: kdhuff@illinois.edu personal e-mail: katyhuff@gmail.com
RESEARCH INTERESTS	Advanced nuclear reactors and fuel cycles, multi-physics simulation, energy systems analysis, scientific computation, nuclear energy policy.	
PHD	University of Wisconsin - Madison, NUCLEAR ENGINEERING • An Integrated Used Fuel Disposition and Generic Repository Model for Fuel Cycle Analysis • Advisor: Professor Paul P.H. Wilson	Aug 2008 – Aug 2013
BA	University of Chicago, PHYSICS • Celestial Gain Calibrations of QUIET Telescope Polarimeters	Aug 2004 – Jun 2008
RESEARCH AND PROFESSIONAL EXPERIENCE	University of Illinois at Urbana-Champaign, Urbana, IL <i>Associate Professor, Nuclear Plasma and Radiological Engineering</i> <i>Affiliate Faculty, National Center for Supercomputing Applications</i> <i>Affiliate Faculty, Computational Science and Engineering</i> Director, Advanced Reactors and Fuel Cycles group.	Nov 2021 – Present Aug 2016 – Present Aug 2018 – Present
	Amazon Web Services, Seattle, WA <i>Amazon Scholar, Data Center Planning and Delivery, Global Energy Team</i> Part-time private advisory consulting.	Aug 2024 – Present
	Office of Nuclear Energy, Department of Energy, Washington, DC <i>Assistant Secretary, Nuclear Energy</i> <i>Senior Advisor to the Secretary, Nuclear Energy</i> <i>Acting Assistant Secretary, Nuclear Energy</i> <i>Principal Deputy Assistant Secretary, Nuclear Energy</i> Presidentially appointed, Senate Confirmed Official leading the Office of Nuclear Energy On extended Unpaid Leave of Absence from the University of Illinois.	May 2022 – May 2024 Jan 2022 – May 2022 May 2021 – Jan 2022 May 2021 – Jan 2022
	University of Illinois at Urbana-Champaign, Urbana, IL <i>Blue Waters Assistant Professor</i> Principal Investigator, Advanced Reactors and Fuel Cycles group.	Aug 2016 – Sep 2021
	University of California - Berkeley, NE Dept., Berkeley, CA <i>Postdoctoral Scholar, Nuclear Science and Security Consortium</i> <i>Data Science Fellow, Berkeley Institute for Data Science</i> Developing computational tools and multiphysics models for advanced reactor safety analysis.	Sep 2013 – Jul 2016 Aug 2014 – Jul 2016
	Argonne National Laboratory, Argonne, IL <i>Laboratory Graduate Research Appointee, Used Fuel Disposition Campaign</i> Developed a used fuel disposition and generic repository computational model.	Jun 2011 – Aug 2013
	University of Wisconsin - Madison, NEEP Dept., Madison, WI <i>Graduate Research Assistant, Computational Nuclear Engineering Research Group</i> Developed and applied CYCLUS, a nuclear fuel cycle systems analysis tool.	Jun 2008 – Aug 2013
	Idaho National Laboratory, Idaho Falls, ID <i>Graduate Research Assistant, Systems Analysis Campaign</i> Developed software functions and requirements for the Fuel Cycle Simulator concept.	Jun – Aug 2010
	Kavli Institute For Cosmological Physics, Chicago, IL <i>Research Assistant, Laboratory for Astrophysics and Space Research</i> Programmed & machined instrumentation. Planned protocol for QUIET polarimeter calibration.	Jan 2005 – Jun 2008

Universidad de Chile, Physics Dept. , Santiago, Chile	Jun – Sep 2006
<i>Research Assistant, Chicago-Chile Research Exchange Program</i>	
Constructed and operated a far-from-equilibrium granular materials experiment.	
Los Alamos Neutron Science Center , Los Alamos, NM	Jun – Sep 2004
<i>Research Assistant, LANSCE-3</i>	May – Aug 2003
Applied digital filtration algorithms and MCNPX models to experimental data.	

HONORS AND AWARDS	NPPE, Students Award for Excellence in Undergraduate Teaching	2017, 2021, 2025
	Fin-Erth Women in Climate Award	2024
	Presidential Citation, American Nuclear Society	2024
	Distinguished Public Service Award, American Nuclear Society (ANS)	2024
	40 Under 40, Nuclear News	2024
	Meritorious Service Award, United States Nuclear Industry Council	2024
	Secretary's Achievement Award, Uranium Strategy Tiger Team, U.S. Department of Energy	2024
	Secretary's Achievement Award, Pathways to Commercial Liftoff Team, U.S. Department of Energy	2023
	Warren K. Sinclair Medal, National Council on Radiation Protection & Measurements	2023
	Presidential Nomination & Senate Confirmation, Assistant Secretary for Nuclear Energy	2022
	Stanley H. Pierce Award, UIUC Engineering Council	2019
	American Nuclear Society, Oestmann Professional Women's Achievement Award	2017
	AE3, Collins Scholars Program Graduate	2017
	UIUC, Teachers Ranked as Excellent	F 2016, S 2020
	American Nuclear Society, Young Member Excellence Award	2016
	National Energy Research Scientific Computing Allocation, Senior Investigator	2015–2016
	Data Science Fellowship, Berkeley Institute for Data Science, UC Berkeley	2014–2016
	Nuclear Science and Security Consortium Postdoctoral Fellowship, UC Berkeley	2013–2016
	DOE Office of Science Laboratory Graduate Appointment, Argonne National Lab	2011–2013
	Roy G Post Foundation Nuclear Waste Management Graduate Scholarship	2011
John Randall Memorial Scholarship, American Nuclear Society FCWMD	2009	
J.A McDeavitt Scholarship, University of Chicago, Chicago, IL	2007–2008	
University Scholar Award, University of Chicago, Chicago, IL	2004–2008	
Los Alamos Distinguished Student Performance Award, Los Alamos National Lab	2004	

GRANTS AWARDED	Nuclear Science and Security Consortium 4	<i>Period:</i> 2026–2031
	<i>Source:</i> DOE-NNSA Office of DNN R&D	<i>Award Total:</i> \$25,000,000
	<i>Role:</i> Consortium Associate Director, UIUC PI, Thrust Area Lead	<i>Huff Allocation:</i> ~\$625,000
	EINSTEIN: Enhanced Integrated Nuclear Systems for Transmutation and Efficient Iso- lation of Nuclides	<i>Period:</i> 2025–2028
	<i>Source:</i> DOE-ARPA-E NEWTON Program	<i>Award Total:</i> \$1,025,405
	<i>Role:</i> Collaborator	<i>Huff Allocation:</i> ~\$250,000
	Nuclear Science and Security Consortium 3¹	<i>Period:</i> 2021–2026
	<i>Source:</i> DOE-NNSA Office of DNN R&D	<i>Award Total:</i> \$25,000,000
	<i>Role:</i> Consortium Co-PI, UIUC PI, Thrust Area Lead	<i>Huff Allocation:</i> \$625,000
	Evaluation of micro-reactor requirements and performance in an existing well-characterized micro-grid¹	<i>Period:</i> 2020–2022
	<i>Source:</i> DOE-NEUP	<i>Award Total:</i> \$800,000
	<i>Role:</i> Co-PI	<i>Huff Allocation:</i> \$265,000
Enabling Load Following Capability in the Transatomic Power MSR¹	<i>Period:</i> 2018–2021	
<i>Source:</i> ARPA - E - MEITNER	<i>Award Total:</i> \$999,694	
<i>Role:</i> Principal Investigator	<i>Huff Allocation:</i> \$205,000	
US Research Software Sustainability Institute (URSSI)	<i>Period:</i> 2017–2018	

¹PI-ship transferred to other leadership in May 2021 corresponding with unpaid leave of absence.

Source: NSF - OAC - SI2 - S2I2 Conceptualization	Award Total: \$499,999
Role: Senior Personnel	Huff Allocation: N/A
Dynamic Transition Analysis with TIMES	Period: 2018–2019
Source: I ² CNER	Award Total: \$76,359
Role: Co-PI	Huff Allocation: \$76,359
Investigation of Agricultural Uses of Nuclear Waste Heat	Period: 2017–2018
Source: Exelon	Award Total: \$151,257
Role: Co-PI	Huff Allocation: \$11,678
Consortium for Verification Technology	Period: 2015–2020
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Co-PI, UIUC PI, CVT Investigator	Huff Allocation: \$347,000
Consortium for Nonproliferation Enabling Capabilities	Period: 2014–2019
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Co-PI, UIUC PI, Thrust Area Lead	Huff Allocation: \$648,000
Collaborative, Open-Source Curriculum Development	Period: 2017–2018
Source: UIUC Strategic Instructional Innovations Program	Award Total: \$19,347
Role: Principal Investigator	Huff Allocation: \$13,000
REU Site: INCLUSION at U. Illinois	Period: 2017–2020
Source: NSF - ACI	Award Total: \$380,036
Role: Senior Personnel	Huff Allocation: N/A
Demand-Driven Cysamore Archetypes	Period: 2016–2019
Source: DOE, NEUP R&D	Award Total: \$800,000
Role: Co-PI	Huff Allocation: \$395,066

- BOOKS
- [1] A. M. Scopatz and **K. D. Huff**. *Effective computation in physics: Field guide to research with python*. O'Reilly Media, Sebastopol, CA, 1st edition, May 2015. URL: <http://shop.oreilly.com/product/0636920033424.do>
- BOOK CHAPTERS
- [2] **K. Huff**. Foreword: Global Need for Nuclear Energy New Directions in. In *Nuclear Energy Innovation and Opportunities in Fission and Fusion for Global Decarbonization*, volume 1. 1st edition edition, Jan. 2026. URL: <https://shop.elsevier.com/books/new-directions-in-nuclear-energy/foss/978-0-443-33471-9>
- [3] S. Gesing, M. Pierce, S. Marru, M. Zentner, **K. D. Huff**, S. Bradley, S. B. Cleveland, S. R. Brandt, R. Ramnath, K. Kee, M. Dahan, B. M. V. Martínez, W. C. Sepulveda, and J. J. S. Mondragón. Science Gateways and AI/ML: How Can Gateway Concepts and Solutions Meet the Needs in Data Science? In *Critical Infrastructure - Modern Approach and New Developments*. IntechOpen, Mar. 2023. URL: <https://www.intechopen.com/chapters/86501>, doi:10.5772/intechopen.110144
- [4] **K. Huff**. Chapter One - Economics of Advanced Reactors and Fuel Cycles. In H. Bindra, editor, *Storage and Hybridization of Nuclear Energy*, volume 1, pages 1–20. Science & Technology Books Elsevier, Inc., Cambridge, MA, United States, 1 edition, Jan. 2019. URL: <http://www.sciencedirect.com/science/article/pii/B9780128139752000016>, doi:10.1016/B978-0-12-813975-2.00001-6
- [5] **K. Huff**. Case Study: Cyclus Project. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017
- [6] **K. Huff**. Lessons Learned. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017
- JOURNAL PUBLICATIONS
- [7] S. G. Dotson, M. Munk, and **K. D. Huff**. Demonstrating the Osier framework for energy system and nuclear fuel cycle optimization. *Annals of Nuclear Energy*, 230:112151, June 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0306454926000393>, doi:10.1016/j.anucene.2026.112151

- [8] A. M. Bachmann, Z. Richter, **K. D. Huff**, and M. Munk. Effects of Uranium Impurities in Down-blended HEU on HTGR Performance. *Nuclear Technology*, 0(0):1–19, Jan. 2026. [eprint: https://doi.org/10.1080/doi:10.1080/00295450.2025.2560690](https://doi.org/10.1080/doi:10.1080/00295450.2025.2560690)
- [9] L. Seifert, S. Shahbazi, S. Richards, M. Munk, and **K. Huff**. Accuracy of the Scaled Flux Method for Spatially Resolved Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors. *Nuclear Science and Engineering*, 0(0):1–13, Sept. 2025. [doi:10.1080/00295639.2025.2568258](https://doi.org/10.1080/00295639.2025.2568258)
- [10] L. Seifert, B. Betzler, W. Wieselquist, M. Jessee, M. Munk, and **K. Huff**. Delayed Neutron Precursor Group Parameter and Spectra Generation from Fast Fission of ²³⁵U in SCALE. *Nuclear Science and Engineering*, 0(0):1–16, Aug. 2025. [doi:10.1080/00295639.2025.2525754](https://doi.org/10.1080/00295639.2025.2525754)
- [11] W. E. Kennedy, R. A. Meserve, K. A. Higley, **K. D. Huff**, C. T. Hanson, M. Ford, D. Schultheisz, T. Smith, B. Kugelmass, A. Abou-Jaoude, J. R. Lovering, J. D. Semancik, G. V. Cullen, J. Cheatham, P. F. Peterson, E. Redmond Li, S. M. Mirsky, M. Mahowald, M. G. Houts, D. Perkins, R. Vaghetto, J. Duhig, and C. J. D. VanHorne-Sealy. National Council on Radiation Protection (NCRP) 2024 annual meeting:advanced and small modular nuclear power reactors. *Journal of Radiological Protection: Official Journal of the Society for Radiological Protection*, 44(4), Oct. 2024. [doi:10.1088/1361-6498/ad7ec4](https://doi.org/10.1088/1361-6498/ad7ec4)
- [12] N. Thiollière, X. Doligez, M. Halasz, G. Krivtchik, I. Merino, B. Mouginot, A. V. Skarbeli, A. Hernandez-Solis, F. Alvarez-Velarde, F. Courtin, H. Druenne, M. Ernoult, **K. Huff**, M. Szieberth, B. Vermeeren, and P. Wilson. Impact of fresh fuel loading management in fuel cycle simulators: A functionality isolation test. *Nuclear Engineering and Design*, 392:111748, June 2022. URL: <https://www.sciencedirect.com/science/article/pii/S0029549322001029>, [doi:10.1016/j.nucengdes.2022.111748](https://doi.org/10.1016/j.nucengdes.2022.111748)
- [13] M. Turkmen, G. J. Y. Chee, and **K. D. Huff**. Machine learning application to single channel design of molten salt reactor. 161:108409, Oct. 2021. [doi:https://doi.org/10.1016/j.anucene.2021.108409](https://doi.org/10.1016/j.anucene.2021.108409)
- [14] A. Chaube, A. Chapman, A. Minami, J. Stubbins, and **K. D. Huff**. The role of current and emerging technologies in meeting Japan’s mid- to long-term carbon reduction goals. *Applied Energy*, 304:117669, Dec. 2021. URL: <https://www.sciencedirect.com/science/article/pii/S0306261921010308>, [doi:10.1016/j.apenergy.2021.117669](https://doi.org/10.1016/j.apenergy.2021.117669)
- [15] A. Chapman, Y. Shigetomi, S. Chandra Karmaker, B. Baran Saha, **K. Huff**, C. Brooks, and J. Stubbins. The cultural dynamics of energy: The impact of lived experience, preference and demographics on future energy policy in the United States. *Energy Research & Social Science*, 80:102231, Oct. 2021. URL: <https://www.sciencedirect.com/science/article/pii/S2214629621003248>, [doi:10.1016/j.erss.2021.102231](https://doi.org/10.1016/j.erss.2021.102231)
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- [17] O. Ashraf, A. Rykhlevskii, G. V. Tikhomirov, and **K. D. Huff**. Strategies for thorium fuel cycle transition in the SD-TMSR. *Annals of Nuclear Energy*, 148, Dec. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0306454920303546>, [doi:10.1016/j.anucene.2020.107656](https://doi.org/10.1016/j.anucene.2020.107656)
- [18] E. A. Miernicki, A. L. Heald, **K. D. Huff**, C. S. Brooks, and A. J. Margenot. Nuclear waste heat use in agriculture: History and opportunities in the United States. *Journal of Cleaner Production*, 267:121918, Sept. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S095965262031965X>, [doi:10.1016/j.jclepro.2020.121918](https://doi.org/10.1016/j.jclepro.2020.121918)
- [19] G. J. Y. Chee, R. E. F. Agosta, J. W. Bae, R. R. Flanagan, A. M. Scopatz, and **K. D. Huff**. Demand-Driven Deployment Capabilities in Cyclus, a Fuel Cycle Simulator. *Nuclear Technology*, 0(0):1–22, July 2020. [eprint: https://doi.org/10.1080/00295450.2020.1753444](https://doi.org/10.1080/00295450.2020.1753444). [doi:10.1080/00295450.2020.1753444](https://doi.org/10.1080/00295450.2020.1753444)
- [20] A. Chaube, A. Chapman, Y. Shigetomi, **K. Huff**, and J. Stubbins. The Role of Hydrogen in Achieving Long Term Japanese Energy System Goals. *Energies*, 13(17):4539, Jan. 2020. Number: 17. URL: <https://www.mdpi.com/1996-1073/13/17/4539>, [doi:10.3390/en13174539](https://doi.org/10.3390/en13174539)

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- [22] O. Ashraf, A. Rykhlevskii, G. Tikhomirov, and **K. D. Huff**. Whole core analysis of the single-fluid double-zone thorium molten salt reactor (SD-TMSR). *Annals of Nuclear Energy*, 137:107–115, Mar. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919306255>, doi: <https://doi.org/10.1016/j.anucene.2019.107115>
- [23] M. Kamuda, J. Zhao, and **K. Huff**. A comparison of machine learning methods for automated gamma-ray spectroscopy. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 954:161385, Feb. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0168900218313779>, doi:10.1016/j.nima.2018.10.063
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- [25] J. W. Bae, J. L. Peterson-Droogh, and **K. D. Huff**. Standardized verification of the Cyclus fuel cycle simulator. *Annals of Nuclear Energy*, 128:288–291, June 2019. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919300179>, doi:10.1016/j.anucene.2019.01.014
- [26] A. Rykhlevskii, J. W. Bae, and **K. D. Huff**. Modeling and simulation of online reprocessing in the thorium-fueled molten salt breeder reactor. *Annals of Nuclear Energy*, 128:366–379, June 2019. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919300350>, doi:10.1016/j.anucene.2019.01.030
- [27] A. Lindsay, G. Ridley, A. Rykhlevskii, and **K. Huff**. Introduction to Moltres: An application for simulation of Molten Salt Reactors. *Annals of Nuclear Energy*, 114:530–540, Apr. 2018. URL: <https://www.sciencedirect.com/science/article/pii/S0306454917304760>, doi:10.1016/j.anucene.2017.12.025
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- [35] M. G. Clerc, P. Cordero, J. Dunstan, **K. D. Huff**, N. Mujica, D. Risso, and G. Varas. Liquid-solid-like transition in quasi-one-dimensional driven granular media. *Nature Physics*, 4(3):249–254, Mar. 2008. URL: <http://dx.doi.org.ezproxy.library.wisc.edu/10.1038/nphys884>, doi:10.1038/nphys884
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- [228] **K. D. Huff.** Doing Our Best: Practices in Scientific Computing. In *PyData Ann Arbor*, Ann Arbor, MI, Feb. 2018. URL: <https://www.youtube.com/watch?v=wkboQuY8Tjg>
- [229] **K. D. Huff.** Why Nuclear? Invited Lecture. In *Olin College of Engineering*, Olin College, Neeham, MA, Oct. 2017
- [230] **K. D. Huff.** Seminar. In *NNSA Nuclear Nonproliferation*, Argonne National Laboratory, Lemont, IL, Sept. 2017
- [231] **K. D. Huff.** Keynote. In *Scientific Python Conference*, Austin, TX, United States, July 2017. URL: <https://scipy2017.scipy.org/>
- [232] **K. D. Huff.** Workforce Transition: How to Succeed-Panel. In *Panel, Young Members Group*, San Francisco, CA, June 2017. URL: <https://epubs.ans.org/?a=40958>

- [233] **K. D. Huff**, J. W. Bae, K. A. Mummah, R. R. Flanagan, and A. M. Scopatz. Current Status of Predictive Transition Capability in Fuel Cycle Simulation. In *Proceedings of Global 2017*, Seoul, South Korea, Sept. 2017
- [234] **K. D. Huff**. Keynote. In *Scientific Python Conference*, Austin, TX, United States, July 2017. URL: <https://scipy2017.scipy.org/>
- [235] **K. D. Huff**. Keynote. In *PyData 2016, Chicago, IL*, Chicago, IL, USA, Aug. 2016. URL: <https://pyvideo.org/pydata-2016/keynote-katy-huff.html>
- [236] **K. D. Huff**. A Computational Future for Science Education. In *Computational, Mathematics, Science & Engineering Seminar, Michigan State University*, East Lansing, MI, USA, Dec. 2015. URL: <https://katyhuff.github.io/2015-11-06-ncsa/>
- [237] **K. D. Huff**. Seminar. In *NNSA Nuclear Nonproliferation, Argonne National Laboratory*, Lemont, IL, Sept. 2017
- [238] **K. D. Huff**. Seminar. In *RPNSD Seminar, Oak Ridge National Laboratory*, Oak Ridge, TN, USA, June 2017
- [239] **K. D. Huff**. Seminar. In *Mechanical and Aerospace Engineering Seminar, UC Davis*, Davis, CA, USA, Apr. 2017
- [240] **K. D. Huff**. Lightning Talk. In *AE3 Lightning Symposium, University of Illinois*, Urbana-Champaign, IL, USA, Mar. 2017
- [241] **K. D. Huff**. Undergraduate Seminar. In *Nuclear, Plasma, & Radiological Engineering Seminar, University of Illinois*, Urbana-Champaign, IL, USA, Feb. 2017
- [242] **K. D. Huff**. Symposium. In *Berkeley Institute for Data Science Symposium, UC Berkeley*, Berkeley, CA, USA, Jan. 2017
- [243] **K. D. Huff**. Seminar. In *Informatics Seminar, University of Illinois*, Urbana-Champaign, IL, USA, Oct. 2016
- [244] **K. D. Huff**. Seminar. In *RPNSD Seminar, Oak Ridge National Laboratory*, Oak Ridge, TN, USA, Mar. 2016
- [245] **K. D. Huff**. Seminar. In *Nuclear Engineering Seminar, University of Tennessee, Knoxville*, Knoxville, TN, USA, Mar. 2016
- [246] **K. D. Huff**. Seminar. In *Nuclear, Plasma & Radiological Engineering Seminar, University of Illinois*, Urbana-Champaign, IL, USA, Dec. 2015
- [247] **K. D. Huff**. Keynote. In *Python in High Performance Computing Workshop, SC15*, Austin, TX, USA, Nov. 2015
- [248] **K. D. Huff**. Colloquium. In *NCSA Colloquium, University of Illinois*, Urbana-Champaign, IL, USA, Nov. 2015
- [249] **K. D. Huff**. Colloquium. In *Nuclear Engineering Colloquium, North Carolina State University*, Raleigh, NC, USA, Oct. 2015
- [250] **K. D. Huff**. Colloquium. In *Nuclear Engineering Colloquium, Texas A&M University*, College Station, TX, USA, Sept. 2015
- [251] **K. D. Huff**. Invited Colloquium. In *Mechanical & Nuclear Engineering Colloquium, Rensselaer Polytechnic Institute*, Troy, NY, USA, Sept. 2015
- [252] **K. D. Huff**. What Can Academia Learn from Open Source? - Panel. In *What Can Academia Learn from Open Source?*, University of Washington, Seattle, WA, USA, Feb. 2015. URL: <https://ti.to/github-events/academia-town-hall->

ENGINEERING
TEACHING

University of Illinois at Urbana-Champaign

DEPT. OF NUCLEAR, PLASMA, AND RADIOLOGICAL ENGINEERING

NPRE 100, Introduction to Nuclear, Plasma, and Radiological Engineering

Fall 2024

Fall 2025

NPRE 247, Modeling Nuclear Energy Systems

Fall 2018

NPRE 412, Nuclear Power Economics and Fuel Management

Fall 2016

Fall 2017

Spring 2020

Spring 2021

Spring 2025

Spring 2026

NPRE 446, Radiation Interactions with Matter I

Fall 2019

NPRE 555, Reactor Theory I

Spring 2018

Fall 2020

Fall 2025

NPRE 560, Reactor Kinetics and Dynamics

Spring 2019

Spring 2021

Fall 2024

GUEST
LECTURES

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Nov 10, 2020

NE 100, Introduction to Nuclear Engineering

Nuclear Fuel Cycle, Advanced Reactors

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Apr 1,3,22, 2015

NE 155, Introduction to Numerical Simulations in Radiation Transport Point Reactor Kinetics, Monte Carlo Methods

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Sep 11, 2014

NE 255, Numerical Simulation in Radiation Transport

Best Practices in Computational Nuclear Engineering

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Apr 1&3, 2013

NE 571, Economic and Environmental Aspects of Nuclear Energy

Nuclear Waste Repository Technology, Policy, and History

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Sep 9&11, 2009

NE 406, Nuclear Reactor Analysis

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Feb 10, 2010

NE 506, Practicum in Monte Carlo Radiation Transport

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

INVITED
SCIENTIFIC
COMPUTING
TEACHING

SciPy Conference, Austin, TX

Jul 6–7, 2015

Introductory Python For Scientific Software

University of Split, Split, Croatia

Sep 8–13, 2014

G-Node Advanced Scientific Programming in Python Summer School

SciPy Conference, Austin, TX

Jun 25, 2013

Version Control and Unit Testing For Scientific Software

University of Chicago, Graduate School , Chicago, IL Computational Literacy Workshop	Jan 12–13, 2013
University of California, Berkeley , Berkeley, CA Department of Statistics Scientific Computing Workshop	Oct 20–21, 2012
Lawrence Berkeley National Laboratory , Berkeley, CA Software Carpentry Python Workshop	Oct 17–18, 2012
International Center for Theoretical Physics , Trieste, Italy UNESCO/IAEA Advanced School on Scientific Software Development	Feb 20–Mar 2, 2012
University of Toronto , Toronto, ON, Canada SciNet Consortium For High Performance Computing Software Carpentry Bootcamp	Nov 7–8, 2011
American Nuclear Society Winter Meeting , Washington, D.C. Young Professionals Congress Hacker Within Scientific Computing Tutorial	Nov 1, 2011
Michigan State University , East Lansing, MI Institute for Cyber Enabled Research (iCER) and BEACON Center THW Bootcamp	Jun 4–5, 2011

SCIENTIFIC
COMPUTING
TEACHING

Berkeley Institute for Data Science , Berkeley, CA Managing Databases in SQL	Jan 14–15, 2015
Berkeley Institute for Data Science , Berkeley, CA Testing for Scientific Software	Jun 4–5, 2015
Lawrence Berkeley National Laboratory , Berkeley, CA Women in Science and Engineering Bootcamp	Apr 14–15, 2014
The University of Chicago , Chicago, IL Software Carpentry Scientific Computing Workshop	Apr 2–3, 2012
The University of Wisconsin , Madison, WI The Hacker Within Software Carpentry Bootcamp	Jan 12–14, 2011
The University of Wisconsin , Madison, WI The Hacker Within Python Bootcamp	Jan 12–14, 2010
The University of Wisconsin , Madison, WI The Hacker Within C++ Bootcamp	Mar 24–31, 2009
The University of Wisconsin , Madison, WI University of Wisconsin, Hacker Within UNIX Bootcamp	Jan 12–15, 2009

POSTDOCTORAL
RESEARCHERS

<u>NAME</u>	<u>DATES</u>	<u>ROLE</u>
Mehmet Turkmen	2019–2020	Advisor
Alexander Lindsay	2016–2017	Advisor

GRADUATE
RESEARCHERS

<u>NAME</u>	<u>DEGREE - YEAR</u>	<u>ROLE</u>
Michael Cheng	MS - 2017	MS Second Reader
Mark Kamuda	MS - 2017	MS Second Reader
Mark Kamuda	PhD - 2019	PhD Advisor
Gregory Westphal	MS - 2019	MS Advisor
Jin Whan Bae	MS - 2019	MS Advisor
Erik Medhurst	MS - 2020	MS Advisor
Andrei Rykhlevskii	PhD - 2020	PhD Advisor
Katherine C. Hepler	PhD - 2020	Dissertation Committee Chair
Alvin Lee	MS - 2020	MS Second Reader
Roberto Fairhurst-Agosta	MS 2020	PhD Advisor
Anshuman Chaube	MS - 2021	PhD Advisor
Gwendolyn Chee	PhD - 2022	PhD Co-Advisor
Amanda Bachmann	PhD - 2023	PhD Co-Advisor

Sun Myung Park	PhD - 2024	PhD Advisor
Nathan Ryan	MS - 2024	MS Advisor
Samuel Dotson	PhD - 205	PhD Advisor
Zoë Richter	PhD - (est. 2026)	PhD Advisor
Luke Seifert	PhD - (est. 2026)	PhD Advisor
Oleksandr Yardas	PhD - (est. 2026)	PhD Advisor
Elijah Capps	PhD - (est. 2029)	PhD Advisor
Liam Pohlmann	PhD - (est. 2029)	PhD Advisor
Samar Abdelkawy	MS - (est. 2026)	MS Advisor

UNDERGRADUATE RESEARCHERS	<u>NAME</u>	<u>DEGREE - YEAR</u>	<u>SCHOLARSHIPS</u>
	Jin Whan Bae	BS - 2017	NPRE Outstanding Undergrad Research ANS Best Student Fuel Cycle Presentation
	Kathryn Mummah	BS - 2017	Roy G. Post Foundation Scholarship ANS FCWMD Randall Scholar
	Eric Riewski	BS - 2017	
	GyuTae Park	BS - (est. 2018)	
	Yukun Tan	BS - (est. 2018)	Students Pushing Innovation
	Lu Kissinger	BS - 2019	
	Xin Wen	BS - 2018	Students Pushing Innovation
	Daniel Chiu	BS - 2020	
	Tyler Kennelly	BS - 2019	
	Bradley Ellis	BS - 2019	
	Adam Pichman	BS - 2019	
	Zoë Richter	BS - 2018	
	Gavin Davis	BS - 2021	
	Kip Kleimenhagen	BS - 2021	
	David Atwater	BS - 2021	
	Nathan Ryan	BS - 2022	
	Anna Balla	BS - 2021	
	Nataly Panczyk	BS - 2024	
	Bella Pequette	BS - 2025	
	Cesar Zambrano	BS - 2025	
	Rhys MacMillan	BS - (est. 2026)	
	Owen Strong	BS - (est. 2028)	
	Matthew Disimone	BS - (est. 2027)	
	Bryan Park	BS - (est. 2027)	
	Thomas Bohnert	BS - (est. 2028)	
	Madeleine Wallman	BS - (est. 2028)	

VISITING RESEARCHERS	<u>NAME</u>	<u>DATES</u>	<u>LEVEL - INSTITUTION</u>
	Gavin Ridey	2017	BS–University of Tennessee, Knoxville
	Aditya Bhosale	2017	BS - IIT, Bombay
	Snehal Chandan	2017	BS - IIT, Bombay
	Eleonora Skrzypek	2019	PhD - Warsaw University of Technology, Poland

SCIENTIFIC COMPUTING SKILLS	Languages	bash/csh, C++, FORTRAN, Perl, Python, XML
	Build Systems	make, CMake, automake
	Databases	HDF5, SQL
	Test Frameworks	CTest, GoogleTest, nose
	Version Control	cvs, git, hg, svn
	Other Tools	Doxygen, Sphinx, GoldSim, L ^A T _E X, Mathematica, MatLab, MCNP, MOOSE

EDITING AND REVIEWING	Editorial Board	<i>Nuclear Technology</i> 2024 – present
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Journal of Open Source Education **2018 – 2021**

Journal of Open Source Software **2016 – 2021**

Nuclear Technology **2018 – 2021**

Nuclear Engineering and Design **2020 – 2021**

Papers in Physics **2020 – 2023**

Proceedings of the SciPy Scientific Python Conference **2013, 2015, & 2017**

Manuscript Referee

Annals of Nuclear Energy

Journal of Nuclear Energy Science and Power Generation Technology

Nuclear Engineering and Design

Nuclear Science and Engineering

Nuclear Technology

Progress in Nuclear Energy

Grant Proposal Referee

Dept. of Energy Nuclear Energy University Programs

Dept. of Energy Technology Commercialization Fund

Blue Waters Fellows Program

Alfred P. Sloan Foundation

Book Proposal Referee

O'Reilly Media

Elsevier

PROFESSIONAL
SERVICE

Member, Argonne Board of Governors Program Review for Nuclear Energy

2025–present

Member, Advocacy Council, Nuclear Matters, Washington D.C.

2025–present

Member, External Affairs Committee, ANS

2025–present

Vice Chair, Mathematics and Computation Division, ANS

2025–present

Advisory Committee, Digital Information Technology, Sloan Foundation

2019–2021

Chair, Nonproliferation and Policy Division, ANS

2020–2021

Executive Committee, Mathematics and Computation Division, ANS

2020–2021

Vice Chair, Nonproliferation and Policy Division, ANS

2019–2020

Chair & Host, Technical Workshop on Fuel Cycle Simulation

2019

Past Chair (*ex officio*), Fuel Cycle & Waste Management Division, ANS

2016–2017

Co-Organizer, Technical Workshop on Fuel Cycle Simulation

2017

Technical Program Committee, IHLRWM Conference

2017

Chair, Fuel Cycle & Waste Management Division, ANS

2016–2017

Vice Chair, Fuel Cycle & Waste Management Division, ANS

2015–2016

Chair, Steering Committee, Software Carpentry Foundation

2014–2015

Secretary–Treasurer, Fuel Cycle & Waste Management Division, ANS

2013–2015

Secretary, Young Members Group, ANS

2013–2014

Technical Program Co-Chair, SciPy, Scientific Python Conference

2013–2014

Member, Next Generation Leadership Committee, Waste Management Symposium

2013–2014

Moderator, Organizer, Panelist, inSCIght Scientific Computing Podcast

2011–2013

Co-Founder, Nuclear Pride, LGBTQA Organization

2011–2013

Co-Founder, Treasurer, President, Hacker Within Scientific Computing Group

2008–2011

Governor, Treasurer, University of Wisconsin ANS student section

2008–2010

DEPARTMENTAL
SERVICE

Graduate Committee, Qualifying Exam Sub-Committee

2024–present

Graduate Committee, Admissions Sub-Committee

Spring 2024

Faculty Advisor, UIUC ANS Student Section

2024–present

Faculty Advisor, UIUC ANS Student Section

2016–2021

Undergraduate Committee,

2019–2021t

Graduate Committee, Qualifying Exam Sub-Committee

2017-2019

Graduate Committee, Admissions Sub-Committee

Spring 2017

	Graduate Committee , Admissions Sub-Committee	Fall 2016
	Advisory Committee ,	2017–2018
	Faculty Search Committee ,	2017–2018
	Faculty Advisor , UIUC WiN Student Section	2017–2018
COLLEGE SERVICE	Member , Courses and Curriculum Committee	2025–present
	Fellow , Education Innovation Fellows	2025–present
	Member , Instructional Facility Working Group	2017–2018
	Selection Committee , Clare Boothe Luce (CBL) Research Scholars	2020–2021
	Member , Engineering IT Governance Education Working Group	2020–2021
	Faculty Mentor , ARISE program	2019–2020
	Member , ENG/TE Liaison Committee	2020–present
	Member , Instructional Facility Working Group	2017–2018
	Faculty Advisor , UIUC CSE The Hacker Within Scientific Computing Group	2016–2017
CAMPUS SERVICE	Senator , U. Illinois at Urbana-Champaign Faculty Senate	2025–present
	Steering Committee Member , Illinois Data Science Initiative	2018
	Hack Mentor , Hack Illinois	2017
CONSULTING	Amazon Web Services Seattle, WA	2024–Present
	<i>Amazon Scholar</i> Nuclear energy due diligence, data center planning and delivery	
	Thomas Edison State University Trenton, NJ	2018–2019
	<i>Subject Matter Expert</i> Institute of Nuclear Power Operations (INPO) Academic Program Review	